CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

51. Consider a binary solution of two volatile liquid components 1 and 2. x_1 and y_1 are the mole fractions of component 1 in liquid and vapour phase, respectively. The slope and intercept of the linear plot of $\frac{1}{x_1}$ vs $\frac{1}{y_1}$ are given respectively as:

(1) $\frac{P_1^0}{P_2^0}, \frac{P_1^0 - P_2^0}{P_2^0}$ (2) $\frac{P_1^0}{P_2^0}, \frac{P_2^0 - P_1^0}{P_2^0}$

(3) $\frac{P_2^0}{P_1^0}, \frac{P_2^0 - P_1^0}{P_2^0}$ (4) $\frac{P_2^0}{P_1^0}, \frac{P_1^0 - P_2^0}{P_2^0}$

Answer (2)

Sol. Given mole fraction of liquid 1 in liquid and vapour phase is x_1 and y_1 respectively.

$P_1 = P_T y_1$ (DLPP)

$P_1 = P_1^0 x_1$ (Raoult's law)

$P_1 = P_T y_1 = P_1^0 x_1$

$$\frac{P_T}{x_1} = \frac{P_1^0}{y_1}$$

$$\frac{P_2^0 + x_1(P_1^0 + P_2^0)}{x_1} = \frac{P_1^0}{y_1}$$

$$\frac{P_2^0}{x_1} + (P_1^0 + P_2^0) = \frac{P_1^0}{y_1}$$

$$\frac{P_2^0}{x_1} = \frac{P_1^0}{y_1} - (P_1^0 + P_2^0)$$

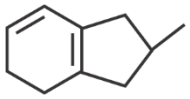
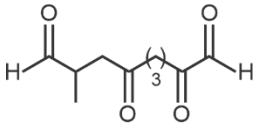
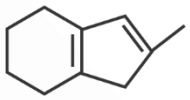
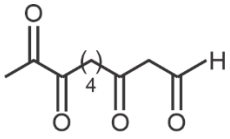
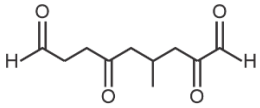
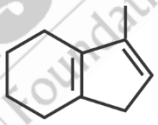
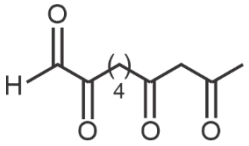
$$\frac{1}{x_1} = \frac{1}{y_2}\left(\frac{P_1^0}{P_2^0}\right) - \frac{(P_1^0 + P_2^0)}{P_2^0}$$

$$\frac{1}{x_1} = \frac{1}{y_2}\left(\frac{P_1^0}{P_2^0}\right) + \frac{(P_2^0 - P_1^0)}{P_2^0}$$

$y = mx + c$

Slope = $\frac{P_1^0}{P_2^0}$; Intercept = $\frac{P_2^0 - P_1^0}{P_2^0}$

52. Match List-I with List-II

List-I (Isomers of $C_{10}H_{14}$)	List-II (Ozonolysis product)
(A) 	(I) 
(B) 	(II) 
(C) 	(III) 
(D) 	(IV) 

Choose the correct answer from the options given below:

(1) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

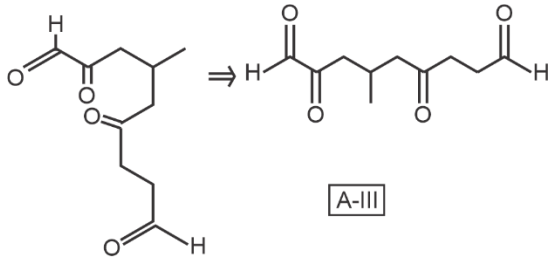
(2) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

(3) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(4) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

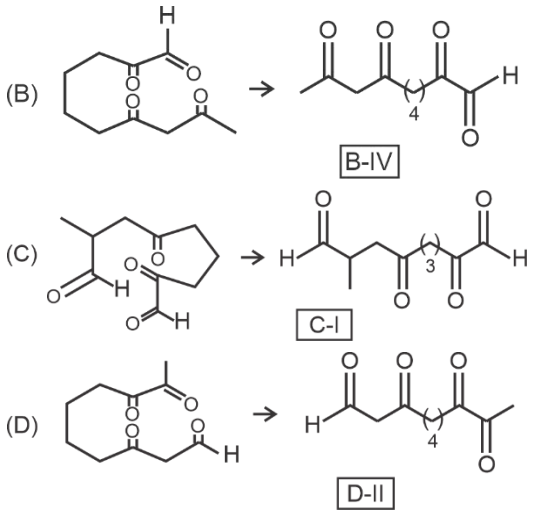
Answer (1)

Sol.

(A) 

A-III





53. Match List-I with List-II

List-I	List-II
(A) Bronze	(I) Cu, Ni
(B) Brass	(II) Fe, Cr, Ni, C
(C) UK silver coin	(III) Cu, Zn
(D) Stainless steel	(IV) Cu, Sn

Choose the **correct** answer from the options given below:

(1) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(2) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

(3) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

(4) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

Answer (1)

Sol.
- Bronze is an alloy of copper and Tin. (A-IV)
- Brass is an alloy of copper and Zinc. (B-III)
- UK Silver coin is an alloy of copper and Nickel. (C-I)
- Stainless steel is an alloy of Fe, Cr, Ni, C. (D-II)

54. Given below are two statements:

Statement (I): For a given shell, the total number of allowed orbitals is given by n^2.

Statement (II): For any subshell, the spatial orientation of the orbitals is given by $-l$ to $+l$ values including zero.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both **Statement I** and **Statement II** are true

(2) **Statement I** is true but **Statement II** is false

(3) **Statement I** is false but **Statement II** is true

(4) Both **Statement I** and **Statement II** are false

Answer (1)

Sol. Statement I is true and statement II is also true.

55. The effect of temperature on spontaneity of reactions are represented as:

	ΔH	ΔS	Temperature	Spontaneity
(A)	+	–	any T	Non-spontaneous
(B)	+	+	low T	Spontaneous
(C)	–	–	low T	Non-spontaneous
(D)	–	+	any T	Spontaneous

The incorrect combinations are:

(1) (A) and (C) only

(2) (B) and (C) only

(3) (A) and (D) only

(4) (B) and (D) only

Answer (2)

Sol. (B) If, ΔH = +, ΔS = +, ΔG is negative at high temperature

Reaction is spontaneous at high temperature

(C) If ΔH = –ve, ΔS = –ve, reaction is non spontaneous at high temperature

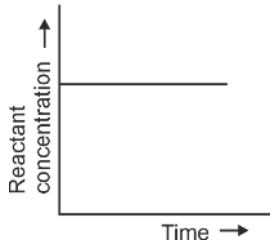
ΔG = –ve

Reaction is spontaneous at low temperature.

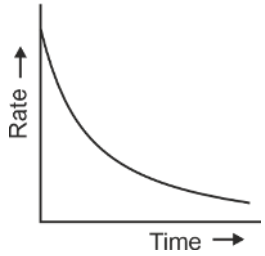


56. Which of the following graphs most appropriately represents a zero-order reaction?

(1)



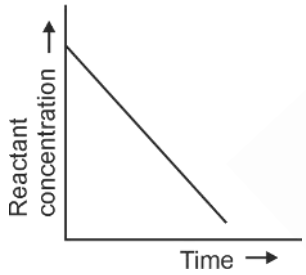
(2)



(3)



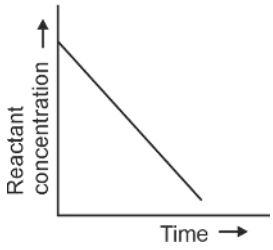
(4)



Answer (4)

Sol. $[A]_t = [A]_0 - kt$

Straight line with negative slope



Graph of reactant concentration versus time for zero order reaction.

57. Given below are two statements:

Statement (I): The boiling points of alcohols and phenols increase with increase in the number of C-atoms.

Statement (II): The boiling points of alcohols and phenols are higher in comparison to other class of compounds such as ethers, haloalkanes.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **Statement I** and **Statement II** are true

(2) Both **Statement I** and **Statement II** are false

(3) **Statement I** is true but **Statement II** is false

(4) **Statement I** is false but **Statement II** is true

Answer (1)

Sol. **Statement I** is correct as boiling point of alcohol phenols increase with increase in the number of C-atoms due to increase in van der Waals forces.

Statement II is correct, since alcohols phenols have intermolecular H-bonding therefore their boiling points are higher in comparison to other class of compounds such as ethers, haloalkanes.

58. Standard electrode potentials for a few half cells are mentioned below:

$E^o_{Cu^{2+}/Cu} = 0.34\ V$, $E^o_{Zn^{2+}/Zn} = -0.76\ V$

$E^o_{Ag^{2+}/Ag} = 0.80\ V$, $E^o_{Mg^{2+}/Mg} = -2.37\ V$

Which one of the following cells gives the most negative value of $\Delta G°$?

(1) $Zn\,|\,Zn^{2+}(1M)\,||\,Ag^{+}(1M)\,|\,Ag$

(2) $Ag\,|\,Ag^{+}(1M)\,||\,Mg^{2+}(1M)\,|\,Mg$

(3) $Zn\,|\,Zn^{2+}(1M)\,||\,Mg^{2+}(1M)\,|\,Mg$

(4) $Cu\,|\,Cu^{2+}(1M)\,||\,Ag^{+}(1M)\,|\,Ag$

Answer (1)

Sol. $E^o_{Cu^{2+}/Cu} = 0.34\ V$, $E^o_{Zn^{2+}/Zn} = -0.76\ V$

$E^o_{Ag^{2+}/Ag} = 0.80\ V$, $E^o_{Mg^{2+}/Mg} = -2.37\ V$



Most negative ΔG° value among the given cells in options is of (1)

$E^\circ = 0.8 + 0.76 = 1.56$

$\therefore \Delta G^\circ = -2 \times F \times 1.56$

$\boxed{\Delta G^\circ = -3.12\ F\ V}$

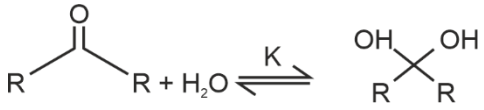
In case of option (2) and (3), E° = –ve

$\therefore$ ΔG° =+ve value,

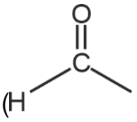
And ΔG° for option (4) will come out to be –0.92 F V, which is less negative than in option (1).

59. Given below are two statements:

Consider the following reaction

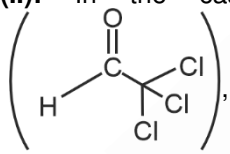


Statement (I): In the case of formaldehyde



, K is about 2280, due to small substituents, hydration is faster.

Statement (II): In the case of trichloro acetaldehyde



, K is about 2000 due to –I effect of –Cl.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both **Statement I** and **Statement II** are true

(2) **Statement I** is false but **Statement II** is true

(3) Both **Statement I** and **Statement II** are false

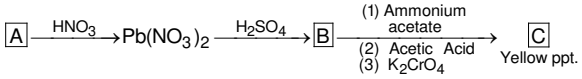
(4) **Statement I** is true but **Statement II** is false

Answer (1)

Sol. Both **Statement I** and **Statement II** are true.

$\boxed{K_{eq} > 1}$, because HCHO and chloral are more electrophilic and equilibrium more lies towards products [Refer: cleyden 1st edition PNo-144 - 145]

60. Identify A, B and C in the given below reaction sequence



(1) $PbCl_2$, $Pb(SO_4)_2$, $PbCrO_4$

(2) PbS, $PbSO_4$, $Pb(CH_3COO)_2$

(3) $PbCl_2$, $PbSO_4$, $PbCrO_4$

(4) PbS, $PbSO_4$, $PbCrO_4$

Answer (4)

Sol. $PbS \xrightarrow{HNO_3} \underset{(B)}{Pb(NO_3)_2} \xrightarrow{H_2SO_4} PbSO_4$

$\xrightarrow[(2)\ \text{Acetic Acid}]{(1)\ \text{Ammonium acetate}} \underset{(C)}{PbCrO_4}$

61. Consider the reaction

$$X_2Y(g) \rightarrow X_2(g) + \frac{1}{2}Y_2(g)$$

The equation representing correct relationship between the degree of dissociation (x) of $X_2Y(g)$ with its equilibrium constant Kp is _____.

Assume x to be very very small.

(1) $x = \sqrt[3]{\frac{2Kp^2}{p}}$ (2) $x = \sqrt[3]{\frac{2\,Kp}{p}}$

(3) $x = \sqrt[3]{\frac{Kp}{2p}}$ (4) $x = \sqrt[3]{\frac{Kp}{p}}$

Answer (1)

Sol. 1 mol

$$\underset{1-x}{X_2Y(g)} \rightarrow \underset{x}{X_2(g)} + \frac{1}{2}\underset{\frac{x}{2}}{Y_2(g)}$$

$$\therefore \quad P_{x_2y} = \frac{1-x}{1+\frac{x}{2}} \times p$$

$$P_{x_2} = \frac{x}{1+\frac{x}{2}} \times p$$



$$P_{y_2} = \frac{x/2}{1+\frac{x}{2}} \times p$$

$$\therefore \quad K_p = \frac{\left(\frac{x}{1+\frac{x}{2}}p\right)\left(\frac{x}{2\left(1+\frac{x}{2}\right)}p\right)^{1/2}}{\left(\frac{1-x}{1+\frac{x}{2}}\right)\times p}$$

$$\therefore \quad K_p = \left(\frac{x}{1-x}\right)\left(\frac{x}{2\left(1+\frac{x}{2}\right)}\right)^{1/2} \times p^{1/2}$$

$\therefore$ x to be very small

$$\therefore \quad K_p = \frac{x^{3/2}}{2^{(1/2)}} \times p^{1/2}$$

$$\therefore \quad x^{3/2} = \frac{K_P \times 2^{1/2}}{p^{1/2}}$$

$$x^3 = \frac{K_p^2 \times 2}{p}$$

$$x = \left(\frac{K_P^2 \times 2}{p}\right)^{1/3}$$

62. Given below are the atomic numbers of some group 14 elements. The atomic number of the element with lowest melting point is:

(1) 82 (2) 6

(3) 50 (4) 14

Answer (3)

Sol. Atomic no. 82 = Lead (Pb)

Atomic no. 6 = Carbon

Atomic no. 50 = Tin (Sn)

Atomic no. 14 = Silicon

The correct order is $\Rightarrow$ C > Si > Pb > Sn

Sn have least melting point

63. pH of water is 7 at 25°C. If water is heated to 80°C., it's pH will :

(1) Increase

(2) Decrease

(3) Remains the same

(4) H^+ concentration increases, OH^- concentration decreases

Answer (2)

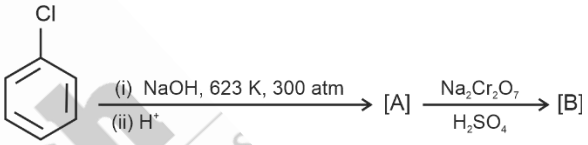
Sol. At 25°C, pure water has pH = 7

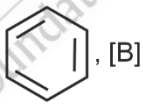
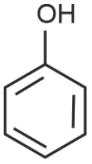
As temperature increased, water molecules dissociate more into hydrogen ions (H^+) and hydroxide ions (OH^-).

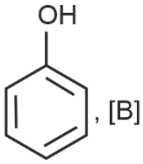
This increased dissociation leads to slightly decrease in pH.

At 80°C, pH ≈ 6.93

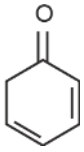
64. Identify the products [A] and [B], respectively in the following reaction :

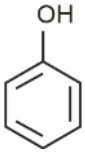


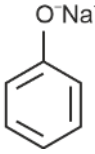
(1) [A]  , [B] 

(2) [A] 

, [B] 

(3) [A]  , [B] 

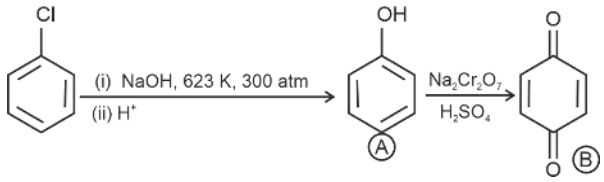
(4) [A] 

, [B] 

Answer (2)



Sol.



65. The α-Helix and β- Pleated sheet structures of protein are associated with its:

(1) Secondary structure (2) Primary structure

(3) Quaternary structure (4) Tertiary structure

Answer (1)

Sol. α-Helix and β-pleated sheets are secondary structures of protein.

66. Given below are two statements about X-ray spectra of elements:

Statement (I) : A plot of $\sqrt{\nu}$ (ν = frequency of X-rays emitted) vs atomic mass is a straight line

Statement (II): A plot of ν (ν = frequency of X-rays emitted) vs atomic number is a straight line

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is true but Statement II is false

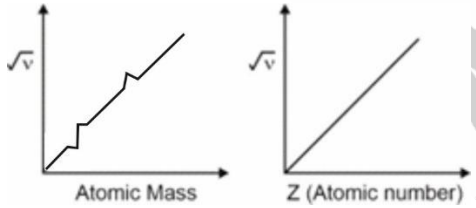
(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are false

(4) Both Statement I and Statement II are true

Answer (3)

Sol.



[Graphs plotted by Henry Moseley]

67. Consider the following reactions

$$K_2Cr_2O_7 \xrightarrow[-H_2O]{KOH} [A] \xrightarrow[-H_2O]{H_2SO_4} [B] + K_2SO_4$$

The products [A] and [B] respectively are:

(1) $K_2Cr(OH)_6$ and Cr_2O_3

(2) K_2CrO_4 and $K_2Cr_2O_7$

(3) K_2CrO_4 and Cr_2O_3

(4) K_2CrO_4 and CrO

Answer (2)

Sol. $K_2Cr_2O_7 \xrightarrow[-H_2O]{KOH} \underset{(A)}{K_2CrO_4} \xrightarrow{H_2SO_4}$

$$\underset{(B)}{K_2Cr_2O_7} + H_2SO_4$$

68. Identify the coordination complexes in which the central metal ion has d^4 configuration

(A) $[FeO_4]^{2-}$

(B) $[Mn(CN)_6]^{3-}$

(C) $[Fe(CN)_6]^{3-}$

(D) $Cr_2(O-\overset{\overset{O}{\|}}{C}-Me)_4(H_2O)_2$

(E) $[NiF_6]^{2-}$

Choose the correct answer from the options given below:

(1) (B) and (D) only (2) (B), (C) and (D) only

(3) (C) and (E) only (4) (A), (B) and (E) only

Answer (1)

Sol. (A) Fe^{6+} = [Ar] $3d^2$

(B) Mn^{3+} = [Ar] $3d^4$

(C) Fe^{3+} = [Ar] $3d^5$

(D) Cr^{2+} = [Ar] $3d^4$

(E) Ni^{4+} = [Ar] $3d^6$

69. When a non-volatile solute is added to the solvent, the vapour pressure of the solvent decreases by 10 mm of Hg. The mole fraction of the solute in the solution is 0.2. What would be the mole fraction of the solvent if decrease in vapour pressure is 20 mm of Hg?

(1) 0.6 (2) 0.2

(3) 0.4 (4) 0.8

Answer (1)

Sol. $\because P^\circ - P \propto x_{solute}$ and $\because 10 \propto 0.2$

$\therefore 20 \propto 0.4$

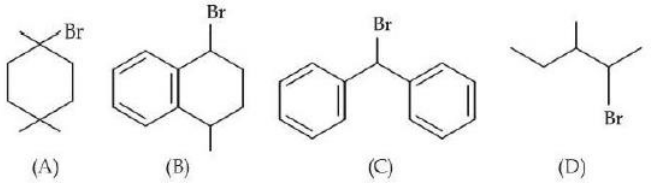
$\therefore x_{solvent} = 1 - x_{solute}$

$= 1 - 0.4$

$x_{solvent} = 0.6$



70. The ascending order of relative rate of solvolysis of following compounds is:



(1) (D) < (B) < (A) < (C)

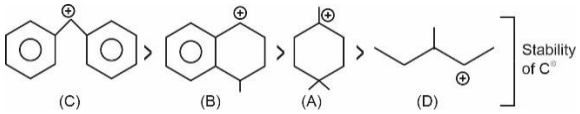
(2) (C) < (D) < (B) < (A)

(3) (C) < (B) < (A) < (D)

(4) (D) < (A) < (B) < (C)

Answer (4)

Sol. Solvolysis or S_N1 order $\propto$ stability of carbocation



SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

71. A compound 'X' absorbs 2 moles of hydrogen and 'X' upon oxidation with $KMnO_4|H^+$ gives

$CH_3-\overset{\overset{O}{\|}}{C}-CH_3$, $CH_3-\overset{\overset{O}{\|}}{C}-OH$ and

$CH_3-\overset{\overset{O}{\|}}{C}-CH_2CH_2-\overset{\overset{O}{\|}}{C}-OH$.

The total number of σ bonds present in the compound 'X' is _____.

Answer (27)

Sol. $CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_2-CH_2-CH=CH-CH_3$ (X) or $CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_2-CH_2-CH=C\overset{CH_3}{\underset{CH_2}{}}$ (X)

$\xrightarrow{KMnO_4/H^+}$

$CH_3-\overset{\overset{O}{\|}}{C}-CH_3 + CH_3-\overset{\overset{O}{\|}}{C}-OH + CH_3-\overset{\overset{O}{\|}}{C}-CH_2CH_2-\overset{\overset{O}{\|}}{C}-OH.$

$\therefore$ 27 σ bonds are present in X

72. 0.01 Mole of an organic compound (X) containing 10% hydrogen, on complete combustion produced 0.9 g H_2O. Molar mass of (X) is _____ g mol^{-1}.

Answer (100)

Sol. Organic compound $\xrightarrow{\text{combustion}}$ $\underset{(0.9\text{ g})}{H_2O}$

$\therefore$ Mole of $H_2O = \frac{0.9}{18} = 0.05$ mol

$\therefore$ Mole of H in $H_2O = 0.05 \times 2 = 0.1$ mole

= Mole of H in 0.01 mol organic compound

$\therefore$ Wt. of H-atom in 0.01 mole of compound

$= 0.1 \times 1$

$= 0.1$ g

$\therefore$ Wt. of H atom in 1 mole compound $= \frac{0.1}{0.01}$

$= 10$ g

$\therefore$ Wt.% of H $= \frac{\text{Wt. of H in one mole compound}}{\text{Molar mass of compound}} \times 100$

$10 = \frac{10}{M} \times 100$

$M = 100$ g/mol

73. The bond dissociation enthalpy of X_2 ΔH°_{bond} calculated from the given data is _____ kJ mol^{-1}. (Nearest integer)

$M^+X^-(s) \rightarrow M^+(g) + X^-(g)$ $\Delta H^\circ_{lattice} = 800$ kJ mol^{-1}

$M(s) \rightarrow M(g)$ $\Delta H^\circ_{sub} = 100$ kJ mol^{-1}

$M(g) \rightarrow M^+(g) + e^-(g)$ $\Delta H^\circ_{i} = 500$ kJ mol^{-1}

$X(g) + e^-(g) \rightarrow X^-(g)$ $\Delta H^\circ_{eg} = -300$ kJ mol^{-1}

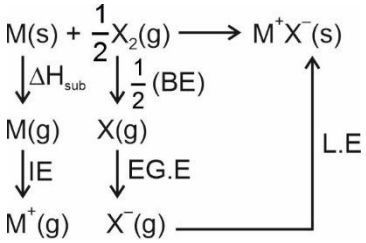
$M(s) + \frac{1}{2}X_2(g) \rightarrow M^+X^-(s)$ $\Delta H^\circ_{f} = -400$ kJ mol^{-1}

[Given M^+X^- is a pure ionic compound and X forms a diatomic molecule X_2 in gaseous state]

Answer (200)



Sol.



$\therefore \quad \Delta H_f(MX) = \Delta H_{Sub}(M) + I.E.(M) + \frac{1}{2}[B.E.(X-X)] + E.GE(X) + L.E.(M)$

$-400 = 100 + 500 + \frac{1}{2}(B.E.) + (-300) + (-800)$

B.E. = 200 kJ/mol

74. When 81.0 g of aluminium is allowed to react with 128.0 g of oxygen gas, the mass of aluminium oxide produced in grams is _____. (Nearest integer)

Given:

Molar mass of Al is 27.0 g mol^{-1}

Molar mass of O is 16.0 g mol^{-1}

Answer (153)

Sol. $4Al + 3O_2 \rightarrow 2Al_2O_3$

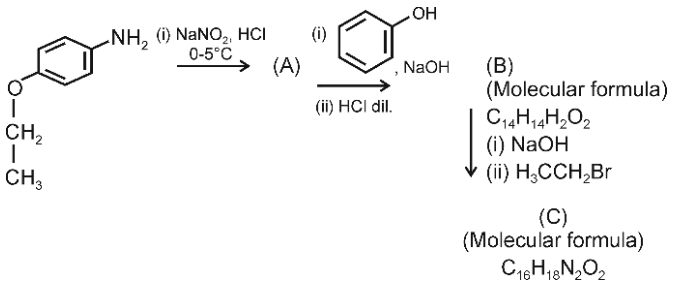
$\frac{81}{27} = 3$ mol $\quad \frac{128}{32} = 4$ mol

Al is Limiting reagent

$\therefore$ Mole of Al_2O_3 formed $= \frac{1}{2} \times 3$ mol

$\therefore$ Wt. of Al_2O_3 formed $= \frac{3}{2} \times 102 = 153$ grams

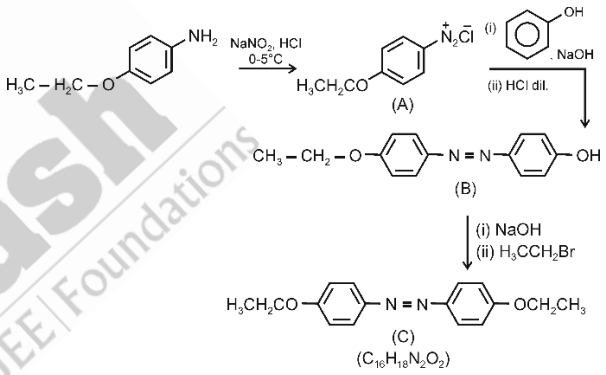
75. Consider the following sequence of reactions.



The total number of sp^3 hybridised carbon atoms in the major product C formed is _____.

Answer (4)

Sol.



Total (4) sp^3 hybridised carbon atoms are present.

❑ ❑ ❑

